Original Article

Drug-Likeness Screening, Toxicity Prediction, and Molecular Docking Study of Phytoconstituents from *Salvia hispanica* against HPr Kinase and Its *In Vitro* Antimicrobial Evaluation

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ABSTRACT

This study investigates the antibacterial potential of *Salvia hispanica* phytoconstituents through inhibition of HPr kinase/phosphatase (PDB ID: 1KNX). Molecular docking was performed using CB-Dock2, and toxicity prediction was assessed via ProTox 2.0. An *in vitro* antibacterial assay evaluated the efficacy of *Salvia hispanica* seed mucilage against *Escherichia coli* by measuring the zone of inhibition at concentrations ranging from 10 to 100 µg/mL. Kaempferol and myricetin exhibited the highest binding affinities (−8.9 kcal/mol), while tocopherol showed the lowest (−7.1 kcal/mol), interacting with key active-site residues including GLU144, GLY147, VAL148 and PHE190. ADME analysis revealed log P values between −1.43 and 8.81 with most compounds classified as mildly toxic. Amikacin demonstrated an estimated LD_{50} of 4,000 mg/kg (toxicity class 5) with potential nephrotoxic, neurotoxic, and pulmonary effects. The antibacterial activity of *Salvia hispanica* seed mucilage was dose-dependent; amikacin showed slightly greater inhibition at lower concentrations (10–40 µg/mL), whereas at higher concentrations (100 µg/mL), both treatments exhibited comparable antibacterial effects, producing a maximum zone of inhibition of 1.9 mm. The findings highlight several promising *Salvia hispanica* bioactive compounds with strong binding affinities, favorable pharmacokinetic properties and confirm the potential of seed mucilage as a natural antibacterial agent.

Introduction

Natural bioactive compounds have gained considerable attention in pharmaceutical research due to their potential therapeutic applications. *Salvia hispanica* (chia) seeds are widely recognized for their rich composition of essential fatty acids, polyphenols, flavonoids, phytosterols, and antioxidants, which collectively contribute to their diverse health benefits. These bioactive constituents play vital roles in various physiological functions and are increasingly explored for novel drug discovery [1,2]. *Salvia hispanica* seeds contain essential fatty acids such as α-linolenic acid, linoleic acid,

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oleic acid, palmitic acid and stearic acid, along with polyphenols including caffeic acid, chlorogenic acid, ferulic acid and gallic acid. Additionally, flavonoids such as quercetin, kaempferol, myricetin, rutin and apigenin, phytosterols like *β*-sitosterol, campesterol, and stigmasterol and vitamin E (tocopherol) further enhance their therapeutic potential [3].

Recent computational studies have highlighted the pharmacological relevance of *Salvia* species. Ononamadu and Seidel employed network pharmacology, molecular docking, and ADME analysis to explore the antidiabetic potential of Salvia officinalis, providing valuable insights for future drug development. Similarly, Didigwu and Nnadi evaluated the drug-likeness, pharmacokinetics, and toxicological profiles of phytotoxic terpenoids, identifying marrubiin and several other compounds as non-toxic, highly bioavailable, and promising drug candidates. Furthermore, Dekebo *et al.* reported significant antibacterial activity, molecular docking, and ADMET profiles of *Salvia rosmarinus* compounds, demonstrating their strong interactions with multiple biological targets and suitability for computational investigations.

Building on these findings, the present study focuses on evaluating bioactive compounds from *Salvia hispanica* seeds using *in silico* approaches. Molecular docking will be performed using the CB-Dock2 platform to predict protein-ligand interactions, utilizing high-quality protein structures obtained from the protein data bank (PDB) for accurate modeling [4,5]. Toxicity prediction will be conducted using the ProTox 2.0 tool, while toxicity prediction and molecular docking were employed to assess the potential interactions of selected compounds with the target protein. Amikacin, a well-established antibiotic, will be used as a standard reference to compare binding efficiency and antibacterial potential.

Through this integrated computational framework, the study aims to identify promising bioactive compounds with strong binding affinities and favorable pharmacokinetic properties, providing valuable insights for future experimental validation and drug development [6,7].

Materials and Methods

Molecular Docking

Ligand Selection: For the present *in silico* investigation, bioactive moieties derived from *Salvia hispanica* were concentrated because synthetic and naturally occurring *Salvia hispanica* seed compounds are well documented in the literature. Additionally, throughout the study, Several reported phytoconstituents of Salvia hispanica were selected as potential ligands for screening [8,9].

Protein of Choice

HPr kinase/phosphatase (PDB ID: 1KNX): Targeting cellular regulatory mechanisms through inhibition of HPr kinase/phosphatase (HPrK/P) represents a promising strategy in antimicrobial drug discovery. HPrK/P is a bifunctional enzyme involved in the phosphorylation and dephosphorylation of HPr and in the regulation of bacterial carbon metabolism. It catalyzes the phosphorylation and dephosphorylation of the histidine-containing protein (HPr), a central component of the phosphotransferase system (PTS) that governs bacterial sugar uptake and utilization. Due to its critical physiological role, HPrK/P has emerged as a potential therapeutic target for developing next-generation antimicrobial agents [10,11].

Natural bioactive compounds are increasingly explored as enzyme inhibitors owing to their structural diversity and pharmacological versatility. *Salvia hispanica* contains numerous phytoconstituents, including polyphenols, flavonoids, and fatty acids with reported antimicrobial, antioxidant, and enzyme-modulating activities. Compounds such as quercetin, kaempferol, caffeic acid, chlorogenic acid and β-sitosterol have demonstrated significant biological functions, making them suitable candidates for enzyme inhibition studies [12].

This study will use molecular docking and toxicity prediction methods to explore how bioactive compounds from *Salvia hispanica* interact with HPr kinase/phosphatase (PDB ID: 1KNX). Tools such as CB-Dock2 will be used for

docking analysis, while ProTox 2.0 will assist in toxicity profiling. The outcomes of this work aim to identify promising lead molecules and offer clues about the potential of *Salvia hispanica* seed constituents as natural antimicrobial agents [13-15].

Molecular Docking: Molecular docking studies are essential for predicting the binding interactions between bioactive molecules and target proteins, thereby supporting the rational design of new drug candidates. In this study, docking simulations were conducted using CB-Dock2 to assess the binding affinities of *Salvia hispanica* derived bioactive compounds toward HPr kinase/phosphatase (PDB ID: 1KNX). The crystallographic structure of the protein was retrieved from the PDB, followed by standard pre-processing steps, including the removal of water molecules, heteroatoms, and unwanted ligands to improve docking precision. Potential binding pockets were identified based on the protein's structural features. The molecular structures of the selected phytoconstituents comprising polyphenols, flavonoids, and fatty acids were drawn using ChemDraw and subsequently energy-minimized to obtain optimized ligand conformations. These optimized compounds were then used as ligands for docking and their binding scores were recorded after generating the appropriate receptor grid for ligand protein interaction analysis [16-18].

Toxicity Prediction: Predictive toxicology plays a crucial role in designing safer and more efficient therapeutic compounds. In this study, ProTox 2.0 software was used to predict the potential toxicological endpoints of *Salvia hispanica* derived ligands. The toxicity parameters evaluated included LD_{50} values, hepatotoxicity, neurotoxicity, nephrotoxicity, respiratory toxicity, and cardiotoxicity. By analyzing these parameters, this study aimed to assess the safety profile of bioactive compounds and determine their potential for further drug development. The molecular structures were submitted to ProTox-II, and the predicted toxicity endpoints were recorded [19,20].

Absorption, Distribution, Metabolism, and Excretion (ADME) Properties: ADME properties are critical factors in evaluating drug-like behavior. The pharmacological efficacy of a compound influences its therapeutic potential, but its metabolic stability and bioavailability also play a role. ADME predictions in this study were performed using ProTox 2.0, which evaluated key pharmacokinetic properties such as partition coefficient (log P), apparent partition coefficient (log D), and intrinsic solubility (log S). Additionally, molecular weight, hydrogen bond donors and acceptors, and other physicochemical parameters were assessed based on Lipinski's Rule of Five, which helps determine drug-likeness. These predictions provided insights into the potential bioavailability and metabolic fate of phytoconstituents from *Salvia hispanica* [21,22].

Evaluation of Antimicrobial Activity

Using the Agar Well Diffusion Method: Using the antibacterial activity of *Salvia hispanica* mucilage powder against *Escherichia coli* was measured by determining the diameter of its inhibition zones [23]. The susceptibility of the chosen organisms to amikacin was evaluated, and the outcomes were noted and documented. The chemical structure of amikacin used as the reference antibiotic is shown in Figure 1.

The *Salvia hispanica* mucilage powder demonstrated potent antibacterial action against *Escherichia coli*. The zone of inhibition (in mm) for both the *Salvia hispanica* mucilage powder (test) and amikacin (standard) at varying concentrations (10 µg/mL to 100 µg/mL) was assessed. The test (*Salvia hispanica* mucilage powder) showed a zone of inhibition of 0.2 mm at 10 µg/mL, which is slightly lower than amikacin's zone of inhibition of 0.5 mm. This indicates that the antimicrobial effect is weaker at this concentration.

The zone of inhibition grew as the amount of *Salvia hispanica* mucilage powder increased. At 100 µg/mL, the test achieved a zone of inhibition of 1.9 mm, which is equivalent to amikacin's performance at the same concentration (1.9 mm). This suggests that at higher concentrations, the test's antimicrobial

effectiveness matches that of the standard antibiotic [24,25].

Results

The molecular docking of selected phytoconstituents from *Salvia hispanica* is listed below in Table 1.

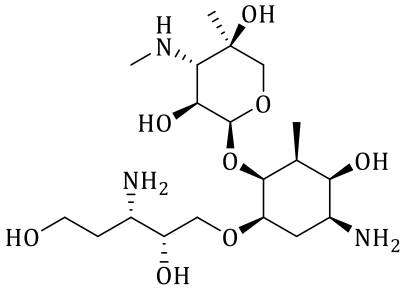


Figure 1: Structure of amikacin as standard for comparison study

Table 1: Molecular docking score and surrounding amino acid sequence of selected phytoconstituent from *Salvia hispanica*

Sr./ No.	Name of phytoconstituent	Molecular docking score	Surrounding amino acid sequence
			Standard
1	Amikacin	-8.8	ARG31VAL32 PRO33 GLY34 LEU35 LYS36 ARG37 SER48 GLN49 ILE50 GLY51 SER52 ILE55 LEU56 GLY57 LYS58 ARG59 GLU60 THR90 LYS91 SER92 PHE115 PHE116 SER117 THR118 GLU119 LEU120 SER121 GLU125
Phytoconstituents			
1	Kaempferol	-8.9	GLU144 GLY147 VAL148 GLY149 ARG184 GLY186 ASN187 ARG188 PHE190 GLU212 LEU217 THR220 LYS221 GLN222 ARG223 THR224 GLU225 ILE226 GLN227 ARG254 GLY257 VAL258 ASP259 PHE284 LYS285 HIS288 SER289
2	Myricetin	-8.9	GLN12 GLU144 PHE146 GLY147 VAL148 GLY149 ARG184 GLY186 ASN187 ARG188 LEU189 PHE190 GLU212 LEU217 THR220 LYS221 GLN222 ARG223 THR224 GLU225 ILE226 GLN227 ARG254 GLY257 VAL258 ASP259 LEU260 LYS285 HIS288 SER289
3	Quercetin	-8.8	LEU165 ILE168 ASN169 LYS170 ASN171 LEU173 PHE190 LYS199 GLU212 PHE214 TYR215 GLY216 LEU217 GLN218 ILE219 THR220 LYS221 GLN222 ARG223 GLU225
4	Chlorogenic acid	-8.6	GLU144 PHE146 GLY147 VAL148 GLY149 ARG184 ASN187 ARG188 PHE190 LYS199 GLU212 LEU217 THR220 LYS221 GLN222 ARG223 THR224 GLU225 ILE226 GLN227 ARG254 GLY257 VAL258 ASP259 PHE284 LYS285 HIS288 SER289
5	Apigenin	-8.6	GLU144 PHE146 GLY147 VAL148 GLY149 VAL175 ARG188 PHE190 LYS198 LYS199 GLU212 LEU217

			THR220 LYS221 GLN222 ARG223 THR224 GLU225 ILE226 GLN227 LYS285 HIS288 SER289
6	Rutin	-8.3	GLU144 GLY147 ILE168 ASN171 LEU173 ARG188 PHE190 LYS199 GLU212 PHE214 TYR215 GLY216 LEU217 GLN218 ILE219 THR220 LYS221 GLN222 ARG223 THR224 GLU225 GLN227 PHE284 LYS285 HIS288 SER289
7	Campesterol	-8.1	PHE146 ILE168 ASN169 LYS170 ASN171 PHE214 TYR215 GLY216 GLN218 ILE219 GLU278 ILE282
8	Stigmasterol	-8.0	PHE146 GLY147 VAL148 GLU181 PHE190 ARG192 ALA193 GLN194 GLU195 LYS198 LYS199 PHE200 ASN210 GLU212 LEU217 THR220 LYS221 GLN222 ARG223 THR224 GLU225 GLN227
9	*β*-Sitosterol	-7.4	PHE146 ILE168 ASN169 LYS170 ASN171 TYR215 GLY216 GLN218 ILE219 GLU275 GLU278 ILE282
10	Tocopherol (vitamin E)	-7.1	PHE146 ILE168 ASN169 LYS170 ASN171 PHE214 TYR215 GLY216 GLN218 ILE219 GLU278 ILE282

Molecular Docking

A comprehensive molecular docking analysis was conducted to evaluate the binding affinity of *Salvia hispanica* derived phytoconstituents with HPr kinase/phosphatase (PDB ID: 1KNX). The docking scores and interactions of these bioactive compounds were compared to identify promising candidates for further investigation. Among the tested compounds, kaempferol and myricetin exhibited the highest binding affinity, each with a docking score of -8.9, interacting with essential amino acids including GLU144, GLY147, VAL148, ARG188, PHE190, LEU217, LYS221, GLU225, and HIS288. Other flavonoids such as quercetin (docking score: -8.8), chlorogenic acid (docking score: -8.6), and apigenin (docking score: -8.6) also demonstrated strong interactions with key active site residues, suggesting their potential inhibitory effects. Additionally, rutin (docking score: -8.3) and sterols like campesterol (docking score: -8.1), stigmasterol (docking score: -8.0) and β-sitosterol (docking score: -7.4) showed moderate binding affinities, indicating their possible role as enzyme inhibitors. Tocopherol (vitamin E) exhibited the lowest docking score (-7.1); yet, its interactions with active site residues suggest a potential role in stabilizing protein-ligand complexes. Amikacin, a well-known antibiotic, was used as a standard to benchmark the efficacy and binding affinity of the *Salvia hispanica*-derived compounds with a docking score of -8.8. Amikacin is used as a standard in both *in silico* and *in vitro* antibacterial studies against HPr kinase because it is a broad-spectrum antibiotic with reproducible antibacterial activity. Although amikacin does not directly inhibit HPr kinase, its strong bactericidal effect validates the experimental system and confirms bacterial susceptibility. In *in vitro* methods, it serves as a positive control, while in *in silico* studies, it provides a reference binding score to compare the activity of phytoconstituents, ensuring comparison and validation of results.

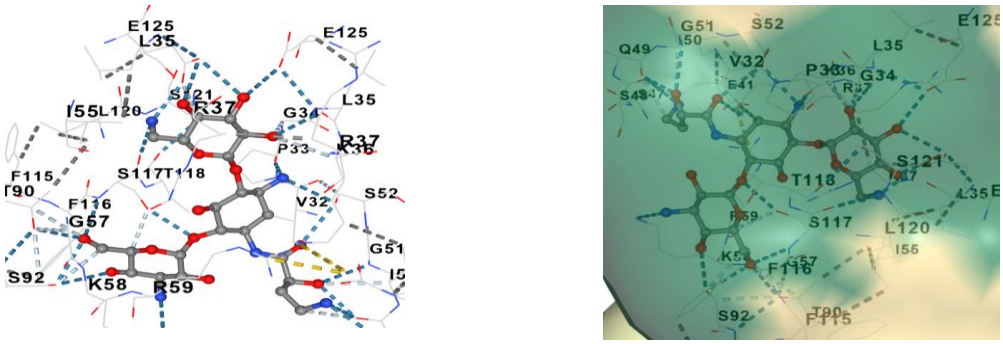
These docking scores indicate that the selected flavonoids, phenolic acids, and sterols from *Salvia hispanica* have promising binding affinities with HPr kinase/phosphatase (PDB ID: 1KNX). Further *in vitro* and *in vivo* studies are necessary to validate their potential as bioactive enzyme inhibitors, paving the way for the development of natural therapeutic agents.

3D Ligand Receptor Interaction Diagram

ADME Properties

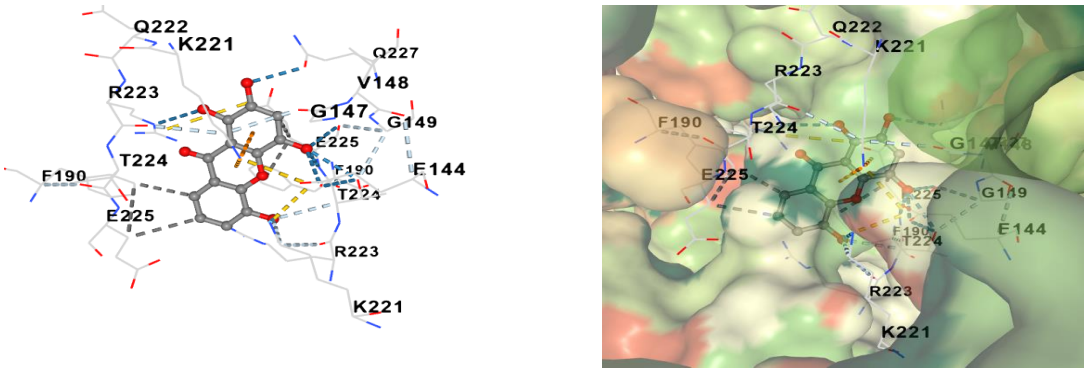
Figures 2a-5a represent the catalytic portion of HPr kinase/phosphatase (PDB ID: 1KNX) proteins with standard amikacin and respective phytoconstituent in the hide receptor interaction was represented in 3D using CB Dock

software. Figures 2b-5b represent the catalytic portion of HPr kinase/phosphatase (PDB ID: 1KNX) proteins with standard amikacin and respective phytoconstituent in drug-receptor interaction was represented in 3D using CB Dock software.



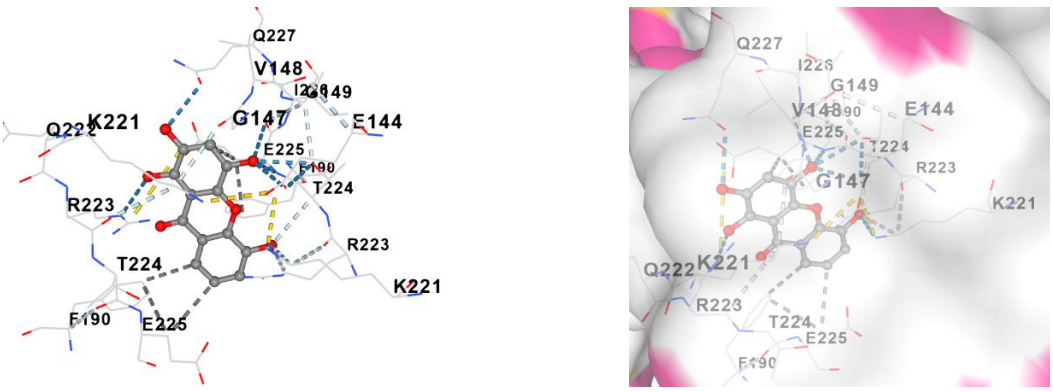
a b

Figure 2: (a) The catalytic portion of HPr kinase/phosphatase (PDB ID: 1KNX) proteins with standard amikacin in the hide receptor interaction was represented in 3D using CB Dock software. (b) The catalytic portion of HPr kinase/phosphatase (PDB ID: 1KNX) protein with standard amikacin in drug-receptor interaction was represented in 3D using CB Dock software.



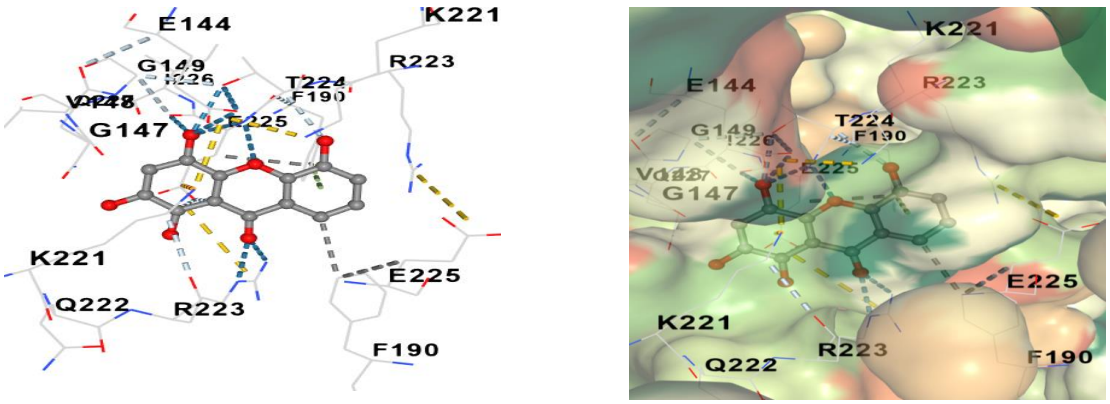
a b

Figure 3: (a) Three-dimensional representation of the interaction of kaempferol with HPr kinase/phosphatase (PDB ID: 1KNX) generated using CB-Dock2. (b) Detailed ligand–receptor interaction of kaempferol with the target protein.



a b

Figure 4: (a) The catalytic portion of HPr kinase/phosphatase (PDB ID: 1KNX) proteins with myricetin in the hide receptor interaction was represented in 3D using CB Dock software. (b) The catalytic portion of HPr kinase/ phosphatase (PDB ID: 1KNX) protein with myricetin in drug-receptor interaction was represented in 3D using CB Dock software.



a b

Figure 5: (a) The catalytic portion of HPr kinase/phosphatase (PDB ID: 1KNX) proteins with quercetin in the hide receptor interaction was represented in 3D using CB Dock software. (b) The catalytic portion of HPr kinase/ phosphatase (PDB ID: 1KNX) protein with quercetin in drug-receptor interaction was represented in 3D using CB Dock software.

The ADME properties of *Salvia hispanica* phytoconstituents were evaluated to determine their pharmacokinetic suitability for drug development. The lipophilicity (log P) values ranged from -1.43 to 8.81, indicating diverse solubility and membrane permeability profiles. Flavonoids like quercetin, and myricetin had moderate log P values (1.77–1.99) and high topological polar surface area (TPSA, >100 Å), which means they are likely to dissolve well in water and be absorbed effectively when taken by mouth. In contrast, sterols such as stigmasterol, demonstrated high log P values (>8), indicative of enhanced lipid solubility but potentially poor aqueous solubility, which may affect systemic absorption and distribution. Table 2 represents the Physicochemical and drug-likeness properties of the phytoconstituents from *Salvia hispanica.* Its pharmacokinetic properties include a molecular weight of 585.6, a high number of hydrogen bond acceptors (18) and donors (13), a TPSA of 331.94 Å, and a log P value of -5.232. A thorough pharmacokinetic and toxicological analysis of *Salvia hispanica* phytoconstituents indicates that flavonoids, phenolic acids, and sterols exhibit advantageous ADME properties, minimal toxicity risk, and potential therapeutic effectiveness. These results endorse their continued exploration for pharmaceutical applications, especially in the development of anti-inflammatory, antioxidant and cardioprotective medications.

Toxicity predictions based on LD_{50} values and toxicity class assessments revealed that most compounds exhibit low to moderate toxicity (Classes 4–6). Essential fatty acids such as α-linolenic acid, linoleic acid, and oleic acid demonstrated the highest LD_{50} values (>10,000 mg/kg, Class 6), indicating a high margin of safety. Conversely, flavonoids like rutin and sterols such as stigmasterol displayed LD_{50} values ranging from 890 to 2,991 mg/kg (Classes 4–5), suggesting the need for further safety evaluations in chronic exposure models. Amikacin, used as a standard in this study, exhibited a predicted LD_{50} value of 4,000 mg/kg (Class 5) and demonstrated potential nephrotoxicity, neurotoxicity, and respiratory toxicity. Table 3 represents the LD_{50} and toxicity class of phytoconstituents from *Salvia hispanica.*

In silico Toxicity Study

Toxicity assessment is a critical step in evaluating the safety profile of bioactive compounds, and computational methods provide an efficient alternative to traditional *in vivo* testing.

Table 2: ADME predictionof phytoconstituents from *Salvia hispanica*

Sr./ No.	Phytoconstituent	Mass	log P	H Bond acceptor	H Bond donor	NRB	TPSA
		Standard					
1	**Amikacin**	585.6	-5.23	18	13	11	331.94
		Phytoconstituents					
1	Alpha-linolenic acid	292.46	6.05	2	1	14	37.3
2	Linoleic acid	266.42	5.49	2	1	13	37.3
3	Oleic acid	268.44	5.72	2	1	14	37.3
4	Palmitic acid	256.42	5.55	2	1	14	37.3
5	Stearic acid	284.48	6.33	2	1	16	37.3
6	Caffeic acid	180.16	1.2	4	3	2	77.76
7	Chlorogenic acid	342.3	-1.33	9	6	5	156.91
8	Quercetin	260.2	1.77	5	4	0	111.13
9	Kaempferol	258.23	-1.43	7	6	1	130.61
10	Myricetin	302.24	1.99	6	5	1	131.36
11	Rutin	442.76	8.81	1	1	9	20.23
12	Apigenin	170.12	0.5	5	4	1	97.99
13	*β*-Sitosterol	258.23	-1.43	7	6	1	130.61
14	Campesterol	302.24	1.99	6	5	1	131.36
15	Stigmasterol	442.76	8.81	1	1	9	20.23
16	Tocopherol (vitamin E)	342.3	-1.33	9	6	5	156.91
17	Ferulic acid	260.2	1.77	5	4	0	111.13
18	Gallic acid	284.48	6.33	2	1	16	37.3

NRB: Number of Rotatable bonds

Table 3: LD_{50} and toxicity class of phytoconstituent from *Salvia hispanica*

Sr./ No.	Name of phytoconstituent	Predicted LD_{50} (mg/kg)	Toxicity class
	Standard		
	Amikacin	4,000	**5**
	Phytoconstituent		
1	Alpha-linolenic acid	10,000	6
2	Linoleic acid	10,000	6
3	Oleic acid	48	2
4	Palmitic acid	900	4
5	Stearic acid	900	4
6	Caffeic acid	2,980	5
7	Chlorogenic acid	4,000	5
8	Quercetin	2,991	4
9	Kaempferol	2,375	5
10	Myricetin	3,919	5
11	Rutin	890	4
12	Apigenin	2,980	5
13	*β*-Sitosterol	2,991	4
14	Campesterol	2,991	4
15	Stigmasterol	2,000	4
16	Tocopherol (vitamin E)	4,582	5
17	Ferulic acid	2,000	4
18	Gallic acid	2,000	4

Using the ProTox 2.0 server, toxicity predictions were performed for *Salvia hispanica* phytoconstituents, focusing on five key toxicity endpoints: hepatotoxicity, neurotoxicity, nephrotoxicity, respiratory toxicity, and cardiotoxicity. Amikacin, used as a standard, exhibited inactive results across all endpoints, except for neurotoxicity, nephrotoxicity, and respiratory toxicity with threshold values of 0.64, 0.93, and 0.80, respectively. Among the analyzed compounds, caffeic acid and chlorogenic acid exhibited hepatotoxic potential, with threshold values of 0.72 and 1.0, respectively, suggesting possible liver-related adverse effects.

Several compounds, including caffeic acid, chlorogenic acid, rutin, apigenin, and gallic acid, displayed neurotoxic activity, indicating a potential risk for central nervous system toxicity. Chlorogenic acid showed the highest neurotoxicity threshold (0.77), necessitating further investigation into its neurological safety. In terms of nephrotoxicity, several compounds, such as caffeic acid, chlorogenic acid, quercetin, kaempferol, myricetin, campesterol, stigmasterol, tocopherol, and ferulic acid, showed active nephrotoxic potential, with values rnging from 0.58 to 0.99. This means that there is a chance that they could harm the kidneys. In terms of respiratory toxicity, flavonoids such as myricetin, rutin, apigenin, campesterol, stigmasterol, tocopherol, ferulic acid, and gallic acid exhibited activity, with rutin and apigenin showing the highest threshold (0.79), highlighting their potential pulmonary risks. Most compounds did not exhibit cardiotoxicity, with kaempferol being the only exception (0.50 thresholds), indicating a low likelihood of cardiovascular toxicity among the tested compounds. Essential fatty acids (alpha-linolenic acid, linoleic acid, and oleic acid) and sterols (β-sitosterol) remained inactive across all toxicity endpoints, supporting their potential as safe pharmaceutical or nutraceutical agents. Table 4 shows *in silico* toxicity prediction of phytoconstituents from *Salvia hispanica.* These computational findings emphasize the importance of toxicity profiling in drug discovery and suggest that while many *Salvia hispanica* constituents hold therapeutic promise, certain compounds may require further preclinical validation to confirm their safety for pharmaceutical applications.

Evaluation of Antimicrobial Activity

The results indicate that the test sample has a dose-dependent increase in its antimicrobial activity. At lower concentrations (10-40 µg/mL), amikacin demonstrated slightly stronger inhibition, but as the concentration increases, both performed similarly, with no significant difference in their ability to inhibit microbial growth at the highest concentration (100 µg/mL). Figure 6 shows Zone of Inhibition of mucilage of *Salvia hispanica* seeds on *Escherichia coli* microorganism.

Discussion

Microbial infections remain a significant health concern globally, necessitating the discovery of effective antimicrobial agents. This study investigates the antimicrobial potential of bioactive compounds derived from *Salvia hispanica*. An *in silico* investigation was conducted utilizing CB-Dock2 to evaluate the binding affinity and interactions of these compounds with HPr kinase/phosphatase (PDB ID: 1KNX).

Ononamadu and Seidel, Exploring the Antidiabetic Potential of *Salvia officinalis* Using Network Pharmacology, Molecular Docking, ADME, and Drug-Likeness Predictions. This study enhances understanding of the antidiabetic activity of *S. officinalis* and provides further insights for future drug discovery purposes [26]. Didigwu and Nnadi, Theoretical Drug-likeness, Pharmacokinetic and Toxicities of Phytotoxic Terpenoids from the Toxic Plants-Phytotoxins. This study identified marrubiin and nine other terpenoids as drug-like, non-toxic, and highly bioavailable with potential for further optimization and development [27]. Dekebo *et al.* describe the antimicrobial activity, docking, and ADMET profiling of *Salvia rosmarinus* compounds for targeting enzymes in cervical cancer. The plant studied shows antimicrobial activities, with the potential for treating cervical cancer, but requires further study [28].

The inhibition zones of *Salvia hispanica* seed mucilage and amikacin at different concentrations are summarized in Table 5.

Table 4: Prediction of *in silico* toxicity of phytoconstituent from *Salvia hispanica*

Sr./ No.	Phytoconstituent	Hepatotoxicity (with threshold value)	Neurotoxicity (with threshold value)	Nephrotoxicity (with threshold value)	Respiratory toxicity (with threshold value)	Cardiotoxicity (with threshold value)
			Standard			
	Amikacin	Inactive (0.97)	Inactive (0.64)	Inactive (0.93)	Inactive (0.80)	Inactive (0.64)
			Phytoconstituent			
1	Alpha-linolenic acid	Inactive (0.55)	Inactive (0.91)	Inactive (0.55)	Inactive (0.84)	Inactive (0.99)
2	Linoleic acid	Inactive (0.55)	Inactive (0.91)	Inactive (0.55)	Inactive (0.84)	Inactive (0.99)
3	Oleic acid	Inactive (0.55)	Inactive (0.91)	Inactive (0.55)	Inactive (0.84)	Inactive (0.99)
4	Palmitic acid	Inactive (0.96)	Inactive (0.97)	Active (0.65)	Inactive (0.85)	Inactive (1.0)
5	Stearic acid	Inactive (0.92)	Inactive (0.93)	Inactive (0.71)	Inactive (0.89)	Inactive (0.97)
6	Caffeic acid	Active (0.72)	Active (0.53)	Active (0.93)	Inactive (0.81)	Inactive (0.71)
7	Chlorogenic acid	Active (1.0)	Active (0.77)	Active (0.99)	Inactive (0.85)	Inactive (0.79)
8	Quercetin	Inactive (0.57)	Inactive (0.83)	Active (0.59)	Inactive (0.59)	Inactive (0.97)
9	Kaempferol	Inactive (0.82)	Inactive (0.88)	Active (0.76)	Inactive (0.67)	Active (0.50)
10	Myricetin	Inactive (0.61)	Inactive (0.88)	Active (0.69)	Active (0.52)	Inactive (0.89)
11	Rutin	Inactive (0.88)	Active (0.52)	Inactive (0.92)	Active (0.79)	Inactive (0.81)
12	Apigenin	Inactive (0.89)	Active (0.56)	Inactive (0.92)	Active (0.79)	Inactive (0.81)
13	β-Sitosterol	Inactive (0.74)	Inactive (0.67)	Inactive (0.79)	Inactive (0.89)	Inactive (0.64)
14	Campesterol	Inactive (0.73)	Inactive (0.90)	Active (0.58)	Active (0.78)	Inactive (0.93)
15	Stigmasterol	Inactive (0.69)	Inactive (0.88)	Active (0.69)	Active (0.63)	Inactive (0.75)
16	Tocopherol (vitamin E)	Inactive (0.69)	Inactive (0.89)	Active (0.62)	Active (0.83)	Inactive (0.99)
17	Ferulic acid	Inactive (0.61)	Inactive (0.88)	Active (0.69)	Active (0.52)	Inactive (0.89)
18	Gallic acid	Inactive (0.88)	Active (0.52)	Inactive (0.92)	Active (0.79)	Inactive (0.81)

**Color Coding Represents*

Table 5: Antimicrobial activity of mucilage of *Salvia hispanica* seeds

Sr./ No.	Particular	Concentration in µg/mL	Diameter of zone of inhibition in mm
1	*Salvia hispanica*seed mucilage (test)	10	0.2
		20	0.6
		40	0.9
		60	1.3
		80	1.4
		100	1.9
2	Amikacin (standard)	10	0.5
		20	0.7
		40	0.8
		60	1.3
		80	1.7
		100	1.9

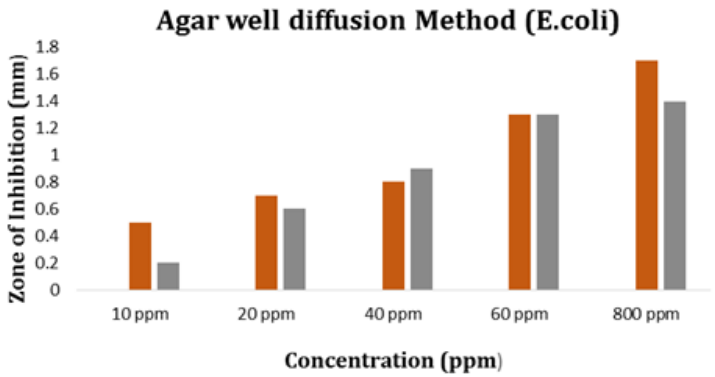


Figure 6: Zone of inhibition of mucilage of *Salvia hispanica* seedson *Escherichia coli* microorganism

Amikacin, a well-established antibiotic, served as a standard reference. The molecular docking analysis demonstrated that kaempferol and myricetin exhibited the highest binding affinity to HPr kinase/phosphatase with docking scores of -8.9, while quercetin, chlorogenic acid, and apigenin also displayed strong binding affinities with scores of -8.8 and -8.6, respectively. The standard compound, Amikacin, showed a docking score of -8.8, underscoring its efficacy. These findings suggest that the selected *Salvia hispanica* compounds have promising interactions with the target enzyme, indicating their potential as antimicrobial agents. The *in vitro* antibacterial assay provided independent evidence of the antibacterial activity of the seed mucilage against *Escherichia coli*; however, it does not directly validate the predicted interaction with HPr kinase/phosphatase. The test sample (*Salvia hispanica* seed mucilage) exhibited a dose-dependent increase in antimicrobial activity against *Escherichia coli*. At lower concentrations (10–40 µg/mL), amikacin showed slightly stronger inhibition, but as the concentration went up, both drugs worked the same way, with no significant difference at the highest concentration (100 µg/mL). The zone of inhibition for *Salvia hispanica* seed mucilage increased from 0.2 mm at 10 µg/mL to 1.9 mm at 100 µg/mL, comparable to amikacin, which showed inhibition zones ranging from 0.5 mm at 10 µg/mL to 1.9 mm at 100 µg/mL. These results suggest that *Salvia hispanica* seed mucilage possesses significant antimicrobial potential at higher concentrations. Further evaluation of the ADME properties revealed diverse solubility and membrane permeability profiles. Flavonoids such as quercetin, kaempferol and myricetin displayed moderate log P values (1.77–1.99) and high topological polar surface area (TPSA>100 Å), suggesting favorable aqueous solubility and oral bioavailability. In contrast, sterols such as β-

sitosterol, stigmasterol and campesterol exhibited high log P values (>8), indicating enhanced lipid solubility but potentially poor aqueous solubility, which could impact systemic absorption and distribution. Toxicity predictions based on LD_{50} values and toxicity class assessments revealed that most compounds exhibit low to moderate toxicity (Classes 4–6). Essential fatty acids such as α-linolenic acid, linoleic acid, and oleic acid showed the highest LD_{50} values (>10,000 mg/kg, Class 6), indicating a high margin of safety. In contrast, flavonoids like rutin and sterols such as stigmasterol had LD_{50} values ranging from 890–2,991 mg/kg (Classes 4–5), necessitating further safety evaluations in chronic exposure models. A comprehensive toxicity assessment was conducted using the ProTox 2.0 server, focusing on hepatotoxicity, neurotoxicity, nephrotoxicity, respiratory toxicity, and cardiotoxicity. Amikacin, the standard, exhibited inactive results across most endpoints except for neurotoxicity, nephrotoxicity, and respiratory toxicity with threshold values of 0.64, 0.93, and 0.80, respectively. Several *Salvia hispanica* compounds, including caffeic acid and chlorogenic acid, displayed hepatotoxic potential with threshold values of 0.72 and 1.0, respectively. Compounds such as caffeic acid, chlorogenic acid, rutin, apigenin, and gallic acid exhibited neurotoxic activity, indicating a potential risk for central nervous system toxicity. Chlorogenic acid had the highest neurotoxicity threshold (0.77), warranting further investigation into its neurological safety. Nephrotoxicity was observed in multiple compounds, including caffeic acid, chlorogenic acid, quercetin, kaempferol, myricetin, campesterol, stigmasterol, tocopherol, and ferulic acid, with values ranging from 0.58 to 0.99, suggesting a possible risk of kidney impairment. In terms of respiratory toxicity, flavonoids such as myricetin, rutin, apigenin, campesterol, stigmasterol, tocopherol, ferulic acid, and gallic acid exhibited activity, with rutin and apigenin showing the highest thresholds (0.79), highlighting their potential pulmonary risks. Most compounds did not exhibit cardiotoxicity, with kaempferol being the only exception (0.50 threshold), indicating a low likelihood of cardiovascular toxicity among the tested compounds. Essential fatty acids (α-linolenic acid, linoleic acid, and oleic acid) and sterols (β-sitosterol) remained inactive across all toxicity endpoints, supporting their potential as safe pharmaceutical or nutraceutical agents. These computational findings emphasize the value of toxicity profiling in drug discovery and suggest that while many *Salvia hispanica* constituents hold therapeutic promise, certain compounds may require further preclinical validation to confirm their safety for pharmaceutical applications. The study emphasizes the prospective use of flavonoids, phenolic acids, and sterols from *Salvia hispanica* as natural antimicrobial agents, necessitating additional *in vitro* and *in vivo* studies to validate their efficacy and safety.

Conclusion

This study provides valuable insights into the antimicrobial potential of *Salvia hispanica* derived bioactive compounds using computational approaches. Molecular docking analysis identified flavonoids and phenolic acids as promising candidates with strong binding affinities toward HPr kinase/phosphatase, indicating their potential role in bacterial growth inhibition. ADME evaluation revealed favorable pharmacokinetic properties, while toxicity profiling suggested an overall acceptable safety profile, although certain compounds warrant further *in vivo* investigation. Future studies should prioritize experimental validation through *in vitro* and *in vivo* antimicrobial assays, exploration of possible synergistic effects with conventional antibiotics and comprehensive safety assessments in biological systems. These findings strengthen the growing evidence supporting natural products as viable sources for antimicrobial drug discovery and highlight their potential for therapeutic development.

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